

SOLUTION

1.EMERGENCY MEDICINE DELIVERY USING AI DRONES

SOLUTION

The proposed solution uses the A* Search algorithm to determine the safest and shortest route for delivering emergency medicines. A* combines the actual travel cost with a heuristic estimate to identify the optimal path. The drone continuously monitors road conditions and weather updates. If a road becomes blocked or weather conditions change, the algorithm recalculates a new route in real time. This approach minimizes delivery time, reduces risk, and ensures reliable emergency medicine delivery in dynamic flood-affected environments.

2.SMART CITY TRAFFIC SIGNAL OPTIMIZATION

SOLUTION

The solution applies a Genetic Algorithm (GA) to optimize traffic signal timings across multiple intersections. The algorithm generates different signal timing combinations and evaluates them based on waiting time, traffic congestion, and fuel consumption. Through selection, crossover, and mutation, the algorithm continuously improves the solutions and adapts to changing traffic conditions. This approach efficiently handles a large search space and provides scalable, real-time traffic management for smart cities.

3.AUTONOMOUS MARS ROVER NAVIGATION

SOLUTION

The solution uses an Online Search Agent that enables the rover to make decisions while exploring unknown terrain. The rover collects information through sensors and cameras, builds an internal map of the environment, and updates it whenever new obstacles or safe paths are detected. If hazards such as rocks or craters are encountered, the rover replans its route to safely reach the destination. This approach allows efficient exploration, safe navigation, and successful sample collection under uncertain conditions.

4.UNIVERSITY EXAM TIMETABLE GENERATION USING CSP

SOLUTION

The proposed solution models the exam scheduling problem as a Constraint Satisfaction Problem (CSP). Each exam is treated as a variable, while available time slots and exam halls form the domains. Constraints such as avoiding overlapping exams, faculty availability, hall capacity, subject order, and special accommodations are enforced during scheduling. A Backtracking Search algorithm combined with Forward Checking efficiently assigns exams to valid time slots by eliminating conflicting choices early. This approach generates a conflict-free and optimized timetable suitable for large universities.

5: STRATEGIC GAME AI USING MINIMAX AND ALPHA-BETA PRUNING

SOLUTION

The solution uses the Minimax algorithm with Alpha-Beta Pruning to develop an intelligent game-playing AI. Minimax evaluates possible future game states by considering both the AI's and the opponent's moves. Alpha-Beta Pruning improves efficiency by eliminating branches of the game tree that cannot influence the final decision, reducing computation time without affecting the optimal result. This enables the AI to make fast, intelligent, and optimal decisions within real-time constraints, providing competitive gameplay against human players.